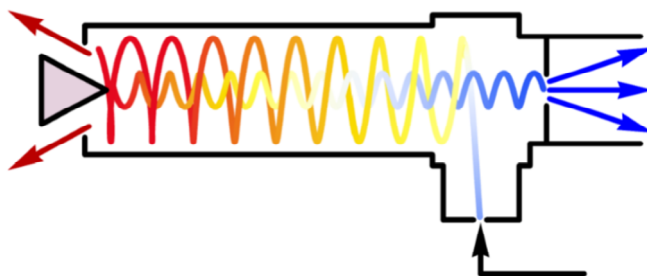


Worksheet: Ranque-Hilsch Vortex Tube

Name(s): _____

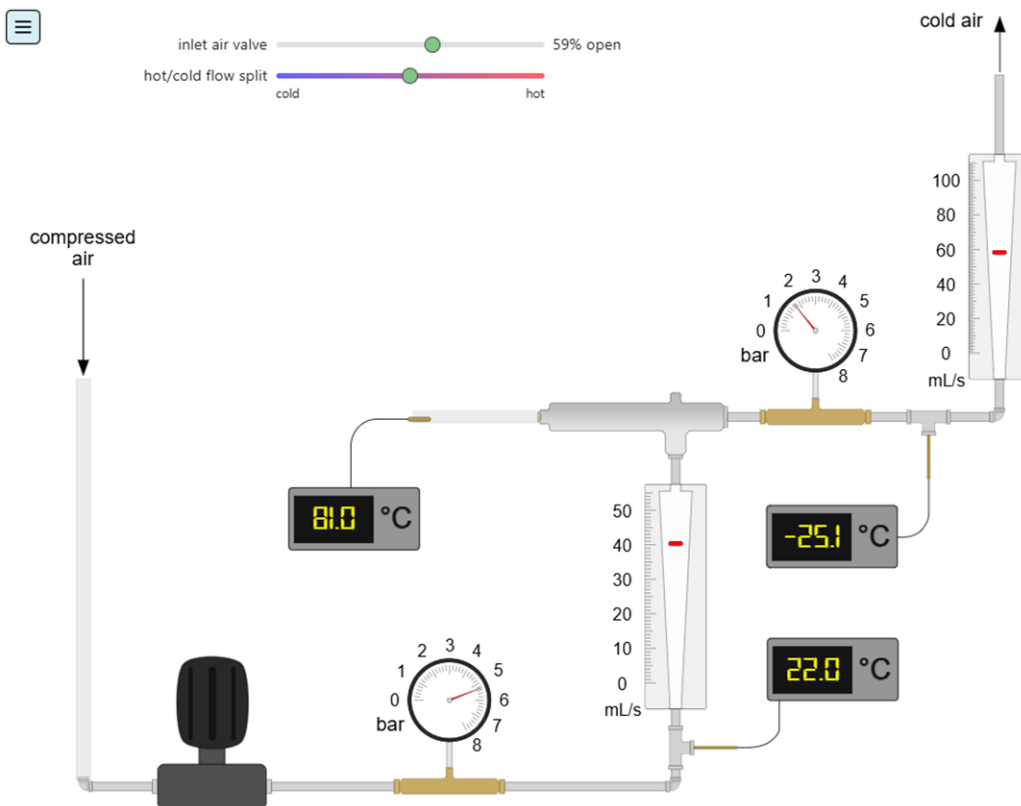
Higher-pressure, room-temperature air is fed to a Ranque-Hilsch vortex tube, which splits the air into hot and cold streams at lower pressure. The fraction of air that goes to the cold side affects how cold the low-temperature side is (or how hot the high-temperature side is). The blue in the figure represents colder temperatures and the red represents hotter temperatures.



Student Learning Objectives

1. Use the ideal gas law to determine molar flow rates.
2. Apply the first law of thermodynamics to a steady-state system.
3. Apply the second law of thermodynamics to a steady-state system.

Equipment



The only moving part in the vortex tube is a valve on the hot side that controls the hot/cold flow split. The hot-side pressure is 1.0 bar, and the cold-side pressure increases as the feed pressure increases. Increasing the inlet air valve opening increases the feed pressure. Rotameters measure the volumetric flow rates for the feed and for the cold air.

Notation

T_f = feed temperature = 22°C

T_c = cold temperature (°C)

T_h = hot temperature (°C)

P_f = feed pressure (bar)

P_c = cold-side pressure (bar)

P_h = hot-side pressure = 1.0 bar

v_f = feed volumetric flow rate (mL/s)

v_c = cold-side volumetric flow rate (mL/s)

$\dot{n}_f$ = feed molar flow rate (mol/s)

$\dot{n}_c$ = cold-side molar flow rate (mol/s)

$\dot{n}_h$ = hot-side molar flow rate (mol/s)

C_p = constant pressure heat capacity = 29.1 J/(mol K)

$\dot{Q}_{cold}$ = cooling rate (J/s)

Questions to answer before running the experiment:

1. Is it physically possible to split a room-temperature stream into a hot and cold stream?
2. Does the total entropy of the system increase, decrease, or not change? Explain why.
3. What are advantages and disadvantages of using a vortex tube over a typical refrigeration system?
4. If more air flows to the hot side of the tube, do you expect the high temperature to increase or decrease? Why?

Running the experiment:

1. Use the slider to open the inlet air valve to about 30% so that compressed air flows to the vortex tube.
2. Adjust the hot/cold flow split so almost all the air flows to the cold outlet.
3. Wait for the system to equilibrate (when the hot and cold temperatures stop changing) and then record the pressures (P_f , P_c) the pressure gauges read absolute pressure), temperatures (T_c , T_h), and volumetric flow rates (v_f , v_c) in Table 1. Note that the rotameter flow rates are read from the top of the float. The hot-side pressure is 1.0 bar. Note that the volumetric flow rate of the cold air is measured at pressures greater than 1.0 bar.
4. Keep the same inlet air valve opening and increase the hot/cold split to about one-third of the range. Wait for the system to equilibrate and record the same measurements in Table 1.
5. Repeat step 4 for a hot/cold split of about two-thirds of the range and for a hot/cold split where most of the air flows to the hot side. Record measurements in Table 1.
6. Move the inlet air valve slider to about 90% open and repeat measurements at the same four hot/cold split settings. Record measurements in Table 1.

Table 1 Temperatures, pressures, and volumetric flow rates ($T_f = 22^\circ\text{C}$, $P_h = 1.0$ bar)

Expt. #	P_f (bar)	v_f (mL/s)	T_c ($^\circ\text{C}$)	P_c (bar)	v_c (mL/s)	T_h ($^\circ\text{C}$)
1						
2						
3						
4						
5						
6						
7						
8						

Analysis:

1. For each measurement, use the ideal gas law to convert volumetric flow rates to molar flow rates for the feed $\dot{n}_f$ and the cold stream $\dot{n}_c$ and record these values in Table 2. A spreadsheet might simplify the calculations.
2. Use the $\dot{n}_f$ and $\dot{n}_c$ values and material balances to calculate the hot stream molar flow rates $\dot{n}_h$ and record those values in Table 2.
3. Calculate the two energy terms ($\dot{n}_c C_p (T_c - T_f)$ and $\dot{n}_h C_p (T_h - T_f)$) in Table 2. Use a heat capacity of 29.1 J/(mol K). Add the two terms together. Do they add to zero?

Table 2 Molar flow rates and energy terms

Expt. #	$\dot{n}_f$ (mol/s)	$\dot{n}_c$ (mol/s)	$\dot{n}_h$ (mol/s)	$\dot{n}_c C_p (T_c - T_f)$ (J/s)	$\dot{n}_h C_p (T_h - T_f)$ (J/s)	Add 2 energy terms(J/s)	$\dot{n}_c C_p (T_f - T_c)$ (J/s)
1							
2							
3							
4							
5							
6							
7							
8							

4. Copy P_f from Table 1 and $\dot{n}_f$, $\dot{n}_c$, and T_c values from Table 2 into Table 3. Calculate the cold fractions ($\frac{\dot{n}_c}{\dot{n}_f}$) and the cooling power ($\dot{Q}_{cold} = \dot{n}_c C_p (T_f - T_c)$) and enter values in Table 3.

Table 3 Cooling power ($T_f = 22^\circ\text{C}$)

Expt. #	P_f (bar)	$\dot{n}_f$ mol/s	$\dot{n}_c$ mol/s	$\frac{\dot{n}_c}{\dot{n}_f}$	T_c	$\dot{n}_c C_p (T_f - T_c)$ (J/s)
1						
2						
3						
4						
5						
6						
7						
8						

Plot T_c versus the cold fraction for each of the feed pressures.

Plot cooling power versus the cold fraction for each of the feed pressures.

What operating conditions maximize cooling power?

5. Calculate the changes in entropy for the cold and hot streams and record in Table 3. Is the total entropy change positive? Should it be? Why?

Table 4 Entropy changes

Expt. #	$\dot{n}_c$ (mol/s)	$\dot{n}_h$ (mol/s)	$\dot{n}_c \left[C_p \ln \left(\frac{T_f + 273}{T_c + 273} \right) + R \ln \left(\frac{P_f}{P_c} \right) \right]$ [J/(K s)]	$\dot{n}_h \left[C_p \ln \left(\frac{T_f + 273}{T_h + 273} \right) + R \ln \left(\frac{P_f}{P_h} \right) \right]$ [J/(K s)]	Add 2 entropy terms
1					
2					
3					
4					
5					
6					
7					
8					

Questions to answer

1. The vortex tube produces one cold stream and one hot stream without a refrigerant or moving parts. Why does this not violate the second law?
2. Which operating condition would you choose to obtain the lowest cold-stream temperature?
3. What is the entropy change of the surroundings? Why?
4. What experimental aspect was ignored in applying the first law to this system?
5. What safety precautions would you observe in conducting this experiment in the laboratory?

Reference

S. Subudhi and M. Sen, "Review of Ranque–Hilsch Vortex Tube Experiments Using Air," *Renewable and Sustainable Energy Reviews*, 52, 2015 pp. 172–178.
[doi:10.1016/j.rser.2015.07.103](https://doi.org/10.1016/j.rser.2015.07.103).